Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/22/2004
PI Name:	Sytkowski, Arthur J. M.D.		
Project Title:	Growth Factor Receptor Function and Cell Differentiation in a Low-Shear Environment		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
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Performance Goal Text:			
	The long-term objective of this research is to use erythropoiesis as a model system for the study of growth factor receptor function, signal transduction and eukaryotic cell growth and differentiation in space. The work is being done initially through a ground-based experimental approach using the low shear environment of the NASA-Rotating Wall Vessel bioreactor (RWV), which has already been proven useful in studies of normal and malignant human cell growth and development. Furthermore, we propose that this research will serve as a basis for later research to be done under the long-term low-shear conditions available in the International Space Station (ISS) under development by NASA. We are studying the effects of reduced hydrodynamic shear stress on erythroid cell surface receptors, receptor-initiated signal transduction and on the assembly of the cytoskeleton in order to elucidate how these effects influence cell growth and differentiation. We are testing the following hypotheses:		

Task Description:	1. Hydrodynamic shear forces affect the interaction of cell surface growth factor receptors with their cognate polypeptide ligands. 2. These changes in receptor function are due, at least in part, to direct effects on receptor distribution, expression and/or protein-protein interactions. 3. Hydrodynamic shear forces on cells also affect downstream signaling events both indirectly, due to effects on the receptor, and directly, through forces that modify signaling molecules within the cell. 4. During growth factor induced cell growth and differentiation, shear forces modulate the assembly of the cytoskeleton, which is an integral part of terminal differentiation. The experiments are designed to elucidate hydrodynamic shear effects on signal transduction, gene expression, and differentiation of red blood cell progenitors. Previous experience with our Epo (erythropoietin) sensitive model erythroid progenitor system and with techniques related to signal transduction and the cytoskeleton and the availability of all required materials and equipment assure that we will be able to achieve the five specific aims of the study within three years. Preliminary data we have generated and previously published related work suggest that we will be able to identify and characterize significant effects of shear on signal transduction, gene expression, and differentiation in red blood cell progenitors. The information obtained through this study is likely to be applicable to the anemia associated with space flight, and is also likely to be applicable to a wide range of other biologic effects of microgravity since the processes we propose to study are basic to the functioning of many cell types.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Changes in cell surface receptor expression and function are a potential means by which cells might sense gravity and hydrodynamic shear (mechanosensing). We now report results of a study carried out on orbit (ISS UF-1) in which an erythroid cell line was induced to differentiate. Rauscher murine erythroleukemia cells, a continuous cell line that can undergo erythropoietin (Epo)- or chemical-induced differentiation similar to normal erythropoiesis, were cultured for 6 days either in microgravity or on earth and then for 3 days in the absence or presence of 50 U Epo/ml or 0.7% dimethyl sulfoxide (DMSO). The cells were then fixed, stored on orbit and returned to Earth for study. Compared to ground-based controls, cells cultured in microgravity exhibited a significantly higher percentage of cluster formation ($p<0.05$) and a greater degree of differentiation (hemoglobinization) ($p<0.01$). Actin content appeared reduced in microgravity, and following Epo or DMSO treatment, there was a more profound loss of actin stress fibers in microgravity. These results are consistent with the hypothesis that erythropoiesis is affected by gravitational forces at the cellular level.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Abstracts for Journals and Proceedings	Xu K, Feldman L, Davis KL, Sytkowski AJ. "Hydrodynamic Shear Affects The Abundance Of Erythropoietin Receptors And Receptor-Mediated Endocytosis." 45th Annual Meeting of the American Society of Hematology, San Diego, California, December 6-9, 2003. Blood 102(11) Part 2:164b, 2003. 45th Annual Meeting of the American Society of Hematology, San Diego, California, December 6-9, 2003. Blood 102(11) Part 2:164b, 2003. , Dec-2003
Abstracts for Journals and Proceedings	Xu K, Holubec KV, Love J, Goodwin TJ, Sytkowski AJ. "Microgravity Affects Cluster Formation, Differentiation, And Cytoskeletal Organization Of Rauscher Murine Erythroleukemia Cells." Annual Meeting of the American Society for Gravitational and Space Biology, Huntsville, Alabama, November 12-15, 2003. Gravitational and Space Biology Bulletin 17(1):38, 2003. Annual Meeting of the American Society for Gravitational and Space Biology, Huntsville, Alabama, November 12-15, 2003. Gravitational and Space Biology Bulletin 17(1):38, 2003. , Nov-2003
Articles in Peer-reviewed Journals	Xu K, Davis KL, Sytkowski AJ. "Measurement of hemoglobin in long-term fixed erythroid cells: application development for cell science experiments in microgravity." Methods Cell Sci. 2003;25(3-4):247-52. PubMed PMID: 15801171 , Sep-2003
Articles in Peer-reviewed Journals	Chen C, Sytkowski AJ. "Erythropoietin regulation of Raf-1 and Mek: evidence for a Ras-independent mechanism." Blood. 2004 Jul 1;104(1):73-80. Epub 2004 Mar 16. https://doi.org/10.1182/blood-2003-04-1340 ; PubMed PMID: 15026317 , Jul-2004
Books/Book Chapters	Sytkowski AJ. "Erythropoietin: Blood, Brain and Beyond." Ed. WILEY-VCH. WILEY-VCH, 2006., Mar-2006
Presentation	Arthur J. Sytkowski "Hydrodynamic Shear and Modeled Microgravity Effects on Erythropoietin Receptor Expression and Function " Annual Meeting of the American Society for Cell Biology: Gravity and Mechanotransduction Cell Signaling Subgroup Dec-2003